

Course Number	0418-111	Course Title	Discrete Mathematics for Computer Science
Credit Hours	3 (3:0:3)	Course Coordinator	Dr. Ludovit Niepel
Compulsory	Yes	Prerequisite	None

Textbook

Title:	Discrete Mathematics and its Applications.
Author:	Kenneth H. Rosen.
Publisher:	McGraw Hill, Latest Edition.

Major Topics Covered in the Course

	Topics	Hours
1	Propositional and predicate logic: Propositions, logical connectives, predicates, quantifiers.	10
2	Fundamental structures: Sets, set operations, power set, Cartesian products; Functions, inverse functions, composition of functions; Sequences, summation.	6
3	Mathematical Reasoning: Methods of proofs, mathematical induction, recursive definitions.	6
4	Counting: Basic counting principles, inclusion-exclusion principle, the Pigeonhole principle, permutations, combinations, Binomial theorem, generalized permutations and combinations.	6
5	Relations Binary relations, relations on sets, composition of relations, reflexive, symmetric, and transitive closure, equivalence relations and partitions, partial order.	5
6	Graphs: Graph terminology, representing graphs, isomorphism, connectivity, trees, applications of trees.	6
7	Exams.	3
	Total.	42

Grading

20%	Midterm 1
20%	Midterm 2
20%	Hmk/quiz/project

40%	Final
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Course Outcomes

Students should be able to:

1. Illustrate by examples the basic terminology of functions, relations, and sets and demonstrate knowledge of their associated operations.
2. Demonstrate in practical applications the use of basic counting principles of permutations, combinations, inclusion/exclusion principle and the pigeonhole principle.
3. Model logic statements arising in algorithm correctness and real-life situations and manipulate them using the formal methods of propositional and predicate logic.
4. Outline basic proofs for theorems using the techniques of - direct proofs, proof by counterexample, proof by contraposition, proof by contradiction, mathematical induction.
5. Illustrate by example basic terminology of graph theory and model problems in computer science using graphs and trees.

Relationship between Course Outcomes and Student Outcomes

Course Outcomes	Student Outcomes					
	1	2	3	4	5	6
1	M					
2	M					
3	H					
4	H					
5	H					